

Dialect Mismatch in Arabic RAG

Summary of findings — target venue: MDPI *Information*

Problem

Arabic knowledge bases are written in Modern Standard Arabic (MSA); users ask questions in dialect. Existing work on this mismatch studies **extractive QA**, where the correct passage is supplied and the model locates an answer span within it. Retrieval is not part of that task.

This study measures the **retrieval stage** of a RAG pipeline — finding the correct passage among thousands — and whether the resulting degradation reaches the answers a user actually receives.

Setup. 3,054-passage Arabic corpus (2,974 Wikipedia passages from 519 Morocco-related articles, plus 80 pilot passages); 500 parallel QA items, each existing as an MSA question and a Moroccan Darija question over the same corpus; hybrid BM25 + dense retrieval; end-to-end generation with LLM-as-judge scoring. All results carry bootstrap 95% confidence intervals.

Finding 1 — Dialectal queries impose a large, robust ranking penalty

Metric	MSA query	Darija query	Gap	95% CI
Recall@1	0.706	0.559	0.147	[0.090, 0.185]
Recall@5	0.923	0.889	0.034	[0.005, 0.085]
MRR	0.798	0.698	0.100	[0.062, 0.131]

Significant for every encoder tested. Critically, the penalty **concentrates at rank 1**: the correct passage is usually still retrieved, it simply stops being ranked first.

Finding 2 — Query normalisation does not fix it, and we can say why

Translating the dialectal query into MSA before retrieval is the intuitive remedy. It was tested as LLM normalisation (two prompt designs), rule-based lexicon substitution, and query expansion, across four encoders and two data subsets. **No configuration improved significantly over the untreated dialectal query.**

Three candidate explanations were falsified by measurement before a fourth was supported:

Query form	Gold-vs-best-competitor margin	Recall@5
MSA	+0.2737	0.913
Darija	+0.0397	0.767
Normalised	+0.0086	0.697

A fluent MSA question matches its target passage well — and also matches many near-neighbour passages well, collapsing the margin that decides rank. A dialectal query is lexically unusual: it matches fewer passages overall, but retains enough distinctive terms that the correct one stays ahead by a wider margin.

Normalisation improves semantic fidelity while reducing ranking discriminability. Absolute overlap with the gold passage rises (0.309 → 0.360); the winning margin falls by a factor of four.

Finding 3 — The ranking penalty does not reach the answers

68 held-out questions, top-5 passages supplied to the generator:

Condition	Gold in context	Correctness	Refusal
MSA query	0.941	0.926	0.015

Condition	Gold in context	Correctness	Refusal
Darija query	0.897	0.882	0.088
Gold passage supplied (oracle)	1.000	0.941	0.029

MSA – Darija correctness: +0.044, 95% CI [-0.029, +0.118] — not significant.

This follows directly from Finding 1. The dialect penalty sits at Recall@1 (0.147) while the Recall@5 gap is only 0.034, and the generator reads the top five passages. The correct passage is nearly always present; its rank is not what the generator responds to.

A RAG pipeline absorbs a moderate ranking penalty. This is a useful negative result: it bounds the practical severity of dialect mismatch in retrieval-based systems.

Finding 4 — Generator prompting outperforms every retrieval-side mitigation

An initial run used a generation prompt with no language constraint, and the generator code-switched into English on dialectal input. Adding one instruction — *answer in Arabic only* — produced this:

	Unconstrained prompt	Arabic-only prompt
Dialectal correctness	0.824	0.882
MSA – Darija gap	+0.132, significant	+0.044, not significant
Code-switching rate	not measured	0.000

One prompt constraint recovered more dialectal answer quality than any retrieval-side intervention tested. It also shows that an apparent end-to-end dialect gap can be an artefact of generator behaviour rather than retrieval failure — a methodological caution for this literature.

Finding 5 — Recall@1 gains do not translate into better answers

Contrastive fine-tuning of the encoder on the parallel MSA/Darija pairs, with passage-level splits and five random seeds:

Metric	Before	After	Gain	Seeds positive
Recall@1	0.559	0.630	+0.070	5/5
MRR	0.698	0.741	+0.044	5/5
Recall@5	0.889	0.897	+0.008	3/5

Stable across 2, 3 and 5 epochs, with no MSA regression (+0.036 Recall@1). **23% of the Recall@1 gap closed.**

End-to-end correctness, however, did not move (+0.000, CI [-0.088, +0.074]) — because Recall@5 was flat and the generator reads top-5.

Retrieval metrics that reward rank-1 ordering can improve without improving a RAG system at all. Where the generator consumes top-k, Recall@k is the metric that matters.

Finding 6 — Author-written benchmarks inflate results through shared style

A fine-tuning run on mixed author-written and Wikipedia data appeared far stronger (+0.110 Recall@5). Split by subset:

Subset	Recall@1 gain	Recall@5 gain	Seeds positive
Author-written	+0.138	+0.110	5/5
Wikipedia	+0.034	-0.002	1/5

The split was by source passage, which prevents memorising individual passages — but not the **author's writing style**, common to every passage they wrote. Removing author-written data entirely *doubled* the effect on real text (+0.034 → +0.070).

Relatedly, questions written while reading their target passage share 0.561 of their tokens with it, against 0.360 for an independent rewrite of the same question.

Passage-level splitting is insufficient when passages share an author. This affects any benchmark built this way, and is worth stating independently of the dialect question.

What this adds up to

Dialect mismatch produces a real and substantial **ranking** penalty in Arabic retrieval, which the intuitive fix does not repair and which we explain mechanistically. At the pipeline level, however, a RAG system largely absorbs it, and the more effective lever proves to be generator prompting rather than retrieval-side mitigation.

Four of the six findings are negative or diagnostic. Each is supported by confidence intervals, and the fine-tuning result by five-seed replication.

Deliverables: a 3,054-passage Moroccan Arabic corpus, a 500-item parallel MSA/Darija benchmark, a six-check validation pipeline, and reproducible retrieval, fine-tuning and generation notebooks.

Limitations and next steps

Scale is the principal weakness: 200 Wikipedia benchmark items, 68 generation evaluations, 7 retrieval-failure cases, one dialect, one domain, one generator. Behaviour on retrieval failure is unresolved — two runs disagreed sharply at $n = 7$, and no claim is made about it.

Highest-value next step: the fine-tuned condition showed no end-to-end effect, so the train/test split it required can be dropped. That raises the generation evaluation from 68 items to all 200 at no additional data cost, and directly narrows the widest confidence intervals in Finding 3.

Then: a second generator to test whether the top-5 robustness in Finding 3 generalises; and an evaluation at top-1 so retrieval failures become frequent enough to analyse.